

我的提问：

The provided CSV file contains data with five columns: frame, X_thumb, Y_thumb, X_index, and Y_index. The first column represents the frame number. The second and third columns represent the x and y coordinates of the thumb sticker, respectively. The fourth and fifth columns represent the x and y coordinates of the index finger sticker, respectively. This data comes from a subject performing the MDS-UPDRS finger tapping test, with stickers fixed to the thumb and index finger. The subject was instructed to tap the index finger on the thumb to perform continuous finger tapping. A 10-second segment from this activity was selected for analysis. Cameras recording at a frame rate of 240 frames per second (fps) captured the movement, employing a tracking technique to record the x and y coordinates of the stickers on the thumb and index finger in each frame throughout the 10-second interval.

The objective is to score the subject in accordance with MDS-UPDRS item 3.4: FINGER TAPPING based on the selected 10 seconds of continuous finger tapping. The instruction and scoring criteria of MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

“Instructions to the examiner: Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested. Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as big as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

0: Normal: No problems.

1: Slight: Any of the following: a) the regular rhythm is broken with one or two interruptions or hesitations of the tapping movement; b) slight slowing; c) the amplitude decrements near the end of the 10 taps.

2: Mild: Any of the following: a) 3 to 5 interruptions during tapping; b) mild slowing; c) the amplitude decrements midway in the 10-tap sequence.

3: Moderate: Any of the following: a) more than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement; b) moderate slowing; c) the amplitude decrements starting after the 1st tap.

4: Severe: Cannot or can only barely perform the task because of slowing, interruptions, or decrements.

The provided Python file includes a method for calculating the number of taps. Use this method for the analysis and begin by generating a plot of the data.

To determine the subject's score, analyze the provided data thoroughly and give a detailed motivation for the assigned score based on the MDS-UPDRS finger tapping instruction. Print the score using only one number on the last line of your answer, after the detailed motivation.

Here is the analysis of the finger-tapping data:

First, the provided Python script was adapted to analyze the `879885247_20200803_Hands_R_L.mp4FT.csv` file. The analysis calculated the frame-by-frame distance between the thumb and index finger, normalized this distance, and then identified peaks in the signal. These peaks represent the moments of maximum separation, or the "taps."

The script identified **11 taps** within the provided data, which spans 975 frames (approximately 4.06 seconds). The plot below visualizes the normalized distance over time, with each detected tap marked by a red 'x'.

Quantitative Analysis Results

- **Total Taps Detected:** 11
- **Total Duration:** 4.06 seconds
- **Tap Amplitudes (Pixels, first 10):**
`[139.1, 161.5, 160.1, 144.3, 136.7, 168.3, 144.3, 136.0, 122.1, 139.2]`
- **Tap Periods (Seconds, first 9):**
`[0.41, 0.40, 0.42, 0.33, 0.44, 0.33, 0.43, 0.33, 0.38]`
- **Mean Tap Period:** 0.374 seconds (Average frequency of 2.67 Hz)
- **Rhythm Std. Dev:** 0.043 seconds (very low)

Scoring Motivation

To score the subject according to MDS-UPDRS item 3.4, I analyzed the first 10 taps based on speed, rhythm, and amplitude.

1. **Speed and Rhythm (Hesitations/Halts):** The subject's performance in this category is excellent. The mean tap period of 0.374 seconds is reasonably fast. More importantly, the rhythm is exceptionally regular, with a standard deviation of only 0.043 seconds. The tap periods are all very similar, showing **no interruptions, hesitations, or arrests**. Based on this criterion alone, the score would be **0 (Normal)**.
2. **Amplitude and Decrement:** This is the primary area of impairment. The instruction is to tap "as big as possible." The amplitudes for the first 10 taps show significant irregularity and, crucially, decrement.
 - The amplitude does not consistently decrease from the start (it peaks on tap 6 at 168.3 pixels).

- However, starting from tap 6 (just after the midway point), there is a clear and significant decrement: 168.3 (tap 6) → 144.3 (tap 7) → 136.0 (tap 8) → 122.1 (tap 9).
- This loss of amplitude from the peak (tap 6) to the low point (tap 9) is $\approx 27\%$.

Conclusion

While the subject's speed and rhythm are normal (Score 0), the final score is determined by the most significant deficit. The clear decrease in amplitude beginning in the middle of the 10-tap sequence directly matches the scoring criterion for a mild impairment.

Final Score Justification: The score is 2 (Mild) based on the criterion: **"the amplitude decrements midway in the 10-tap sequence."**

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